

Table 1: Error source summary

Source	Observable	Description	Rel. range (%)
Camera calibration	T_{avg}	Linear rescaling; $\pm 5^\circ\text{C}$ uncertainty at reference point	0.711–1.42
Timing delay	T_{avg}	1-s readout lag propagated via cooling rate $ dT/dt $	0.00841–0.0985
Combined temperature	T_{avg}	Calibration + timing added in quadrature	0.711–1.42
Pixel measurement	V_{pp}	Manual pixel height reading of signal amplitude	0.814–3.85
Calibration bar	V_{pp}	Pixel uncertainty on mV scale bar; sets px $\rightarrow$ mV conversion error	1.13–1.9
T_2 fit (statistical)	T_2	Curve-fit covariance from 3-pt ($\tau=10,16,18$ ms) exponential fit	24–51
T_2 fit (systematic)	T_2	Shift between 2-pt and 3-pt fit; sensitivity to τ point selection	52.7–92.6